Cybernetics, Reflexivity and Second-Order Science

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Introduction

1. In this article, I study the questions "What is cybernetics?" and "What is science?" I examine these questions in the form of reflexivity. I shall explain what is meant by a reflexive domain and in the process expand the concept of cybernetics. Cybernetics is concerned with circularity, a circularity that includes the observer (or operator) in the system. The observer is actively in the system. Nevertheless, domains with such circularity remain amenable to rational study. In cybernetics, one attempts to be fully aware of the context of any situation. This means that a reflection on the context, and an inclusion of that awareness of context into the context is always present. What is being evoked is a different sense of the rational than that offered by traditional science. In the face of the circularity of context and observer it is still possible to explore and come to agreements that have every appearance of being scientific facts. I shall describe how it comes about that an arena generated by dynamic interactions and shifting relationships can become a world just like ours, subject to exploration, invention and discovery. One may say that from this point of view, objectivity is an emergent phenomenon! The body of the article is formed in a series of short, numbered paragraphs that I hope will let the reader explore and compare the ideas as they are articulated. The main ideas have already been expressed in this paragraph, but extensions and ways to speak and express these notions occur in the expanded form of the article itself.

2. As cybernetics grew, this notion of circular domains, in which cybernetics occurs, also grew. There are areas of logic and mathematics that have a good correspondence with the sort of circularity that cybernetics needs. I shall speak of reflexive domains.

3. A *reflexive domain* is an abstract description of a conversational domain in which cybernetics can occur. Each participant in the reflexive domain is also an actor who transforms that domain. In full reflexivity, each participant is entirely determined by how he or she acts in the domain, and the domain is entirely determined by its participants. I write $D=[D,D]$ to denote this reflexivity of a domain D (Kauffman 1987, 2001, 2004, 2005, 2009, 2012a, 2012b, 2015; Scott 1971; Varela 1979).[1] The symbol $[D,D]$ denotes all the

1. These references deal with both lambda calculus and the notion of reflexive domains. In this listing I have indicated only those references that explicitly or implicitly deal with reflexive do-

available transformations of the domain D to itself. The equation says that D is identical to the processes that transform it. The point to note is that if a transformation T is defined by the equation $T(D)=[D,D]$, then the transformation T applied to a domain D reveals all the processes that can transform D into itself. *A reflexive domain is itself an eigenform* (see below): $D=T(D)=[D,D]$. Note that a reflexive domain is a context for action, and when we say that the domain D is itself an eigenform, we stand back momentarily from the domain into a larger context that can include it. This means that no domain, even a reflexive domain, is the end of our deliberations. Each domain can be transcended to a new and larger domain. The process is endless and is the source of all our constructions and considerations.

4. An eigenform is a fixed point for a transformation. In the context of this article, an arbitrary transformation is allowed in any mathematical domain with a fixed point either in that domain or in some extension of that domain. This usage is an extension of some technical uses of the term that are special cases of this notion. In the next paragraph, I give a specific example of how an eigenform can arise as the fixed point of a transformation that is simple and syntactical. For our purposes an eigenform is the analog of an eigenvector in analysis or linear algebra, but it is much more general and includes the fixed points that occur in reflexive domains, as will be explained below.

5. I define $T(x)=[x]$. Then I can apply T again and again to an arbitrary x as shown below:

$$\begin{array}{c} x \\ [x] \\ [[x]] \\ [[[x]]] \\ [[[[x]]]] \\ [[[[[x]]]]] \\ \ldots \end{array}$$

6. If you do this for a long time it begins to look like

$$\mathrm{E}=[[[[[[[[[[\ldots]]]]]]]]]]$$

and this expression has the form of something that does not change if you put one more set of brackets around it. Thus E (above) is an eigenform for the transformation T where

$$T(x)=[x].$$

7. The entity E appears in our perception due to the recursive action of the transformation, and it is a consequence of how one deals with E, seeing it as invariant under T, that makes it into an object for our perception. That E appears as an object is part and parcel of being an eigenform. Thus Heinz von Foerster spoke of eigenforms in the phrase "objects as tokens for eigenbehaviors" (Foerster 2003a). Heinz, in a wonderful turn of perception, turned the mathematical idea on its head. He pointed out that ordinary objects are tokens for eigenbehaviors. *Ordinary objects are invariances of processes performed in the space of our experience.* The "space of our experience" is the context in which we have our experience and it is the experience itself. I make objects by finding fixed points in the recursion of

mains. A reflexive domain is a particular situation where lambda calculus applies.

my interactions. Eigenforms are a touchstone for the relationship of circular and recursive processes and the ground of our apparent worlds of perception. This point of view should be compared with that of Humberto Maturana and Francisco Varela (1987), where they see the construction of conversational domains through the coordination of coordinations of actions of organisms in the course of their evolution.

8. With this in mind, look again at the defining property of the reflexive domain $D, D=[D,D]$, and understand that *the domain itself is an eigenform*.

9. This structural observation has consequences for the applications of cybernetics and applications to the epistemology of second-order science (Kauffman 2015; Müller & Riegler 2014; Umpleby 2014). For if science is to be performed in a reflexive domain, then one must recognize the actions of the persons in the domain. Persons and their actions are not separate. If an action is a scientific theory about the domain, then this theory becomes a (new) transformation of the domain. Theory inevitably affects the ground that it studies. Furthermore, the fact that an entire domain can be seen as an eigenform suggests that one can be an observer of that domain in a wider view of the landscape. Thus physics can be seen as a reflexive domain and one can take a meta-scientific view, allowing physics itself to be one of the objects of a larger domain of in which it (physical science) is one of the eigenforms.

10. How then, do physical and natural science manage to obtain their apparently objective results? The answer lies in circularity and eigenform. Traditional science searches only for those actions that are independent of the observers (persons) involved. This means that traditional science asks to work within a particular subset of available eigenforms. Once this is realized, one can begin to see how to widen scientific operations.

11. Exploration of this theme will proceed in other papers, but the point I want to make here is that the initial place to begin cybernetics and second-order science is in the recognition of circularity and reflexivity.

12. In this article, I shall show that there is no definition of cybernetics that is not a circular definition. This result should be a cause for celebration, for cybernetics is a study of circularity and the fact that it requires a circular definition shows that cybernetics is fundamentally not separate from circularity.

13. The next section discusses definitions and the structure of circular definitions followed by a section showing that basic notions of mathematics would require circular definitions but that mathematics, allowing as it does undefined terms, can give the appearance of avoiding circularity. I continue with a section providing the proof that cybernetics, if it has definitions, must have circular definitions.

14. The subsequent section is a discussion that leads outward to the structure of science and the meaning of eigenform and reflexive domains. I show that our concept of reflexive domain is itself an eigenform and that the central, circular concept of eigenform underlies and overlies all notions of reflexivity. In the Conclusion, I discuss science and second-order science directly. It is my thesis that all science, all attempts to find knowledge, are faithfully modeled by the search for eigenforms in a reflexive domain. A conventionally successful science will find such forms and point to them as the objective results of that science. But a wider look at the situation will reveal that the larger landscape of the reflexive domain has been significantly influenced by these theories. The world has changed as a result of the scientific activity. There is no inviolate ground. The path is being constructed in the act of making it. And yet there are beautiful and important eigenforms to

be found. The quest of science becomes larger and more romantic and more intentional in the embracing of the second-order. I imagine worlds and bring them into being.

Distinction and circularity

15. How do definitions come about?

16. A definition of a term, as it is usually taken, defines that term in terms of other terms. For example, I may say that "a hammer is a device for pounding nails into wood" or "a straight line segment is the shortest distance been two points in a plane space." Definitions in this form are to be found in the dictionary. Some words are genuinely difficult to define. Consider what happens when I look up the world "distinction" in a dictionary. I find that the word distinction is defined as a marking off, an indication of a difference, a discrimination or a distinguishing quality or characteristic.

17. If you already know the meaning of distinction, the dictionary entry can be useful in pinning down some subtleties of usage of the word. But if you did not know the meaning of the word, then you would only find synonyms for distinction in the definition of distinction.

18. The dictionary has engaged in circularity in defining distinction.

19. According to the dictionary:

1. A distinction is a marking off of a difference.
2. A difference is the making of a distinction.

20. This is a circular definition. Each term is defined in terms of the other term. I have been taught that circular definitions are wrong and should be avoided. And yet our dictionaries routinely use circular definitions for the most fundamental terms and concepts such as distinction. I say that the circularity of definitions of distinction is a cause for celebration. For it cannot be otherwise.

21. What is a definition? In order for a definition to be effective it must make a distinction. It must carve a niche in the world of our words. In order for a definition to tell us without circularity what is a distinction, that definition must distinguish distinguishing. This cannot be accomplished without circularity.

Theorem 1. There is no definition of distinction that is not circular.

Proof. If there were a definition of distinction, this would be a distinction that characterizes distinction. Thus the definition would be circular, using the concept of distinction to define distinction. Hence there is no definition of distinction that is not circular. *QED*

22. This theorem does not tell me that I cannot understand distinction. It tells me that such understanding is necessarily a matter of experience. Make a distinction. Draw a circle in a tide-flattened stretch of sand. Bake a cake. Prove a theorem. Sing a song. Attempt to understand understanding. Live a life. Distinction transcends closed worlds of words, and moves into the worlds of feeling and action.

Mathematics

23. Mathematics is based on the idea of a distinction and mathematics is usually regarded as a gold standard for precision. No one said that distinctions could not have precision. I said that they must have circularity, but circularity does not preclude precision.

24. Take numbers, positive integer numbers such as 1, 2, 3. We all seem to agree how to treat them and work with them and they are precise. If I say that I have one apple, you know what I mean. But I may ask what is the number 2? This is a question at a different level.

25. The number 2 is represented by the set TWO = { { }, { { } } } whose members are the two sets { } and { { } }. The first set is empty and the second set has as its member the empty set. Thus these two sets are distinct, and so TWO is a set with two distinct members. TWO represents the number 2.

26. I produced the set TWO and checked that it had two distinct members. So I already knew what two was and used it in the definition of TWO. But TWO is not two! TWO is a set and it is like a ruler. TWO is *the mathematician's standard for the number two.* If you ask a mathematician "Are there two sheep in that field?", he pulls out the set TWO from his waistcoat and compares it with the sheep in the field. He then says to you "I have attempted to make a one-to-one correspondence between the sheep in the field and my set TWO. I discover that there are more sheep in the field than there are members of TWO. Therefore I tell you that there are more than two sheep in that field."

27. I do not need a definition for 2. If someone wants to know if there are 2 things about, I just take out my ruler and check. But could I give a definition of 2?

28. Bertrand Russell gave a definition of number. He writes "[…] a number of a class [is] the class of all classes similar to the given class" (Russell 1938: 115).

29. Thus Russell would say that 2 is the collection of all (possible) pairs. Russell's definition is a deep insight, but it is very wide. How am I to know about all pairs? Let us keep thinking about Russell's idea but leave it aside for now.

30. What if I said that 2 is the set { { }, { { } } }? That will not do. I could call this set TWO, but I cannot call it 2 because *2 is the concept of a pairing* and the set is just a set. So I use the set as a definite example of a pairing, and then I say that something has 2 elements if they can be matched with this set.

31. Now there is an even deeper problem. In defining 2 via this set and the idea of matching, I use the idea of two for the matching. Two things are matched if you put them in correspondence with one another, and so I use the concept of 2 in defining the concept of 2. Matching and pairing mean the same. As I see these features of twoness, I get a precise idea about the concept 2.

32. Precision of concept is something that one can explore, even though there is no way to avoid circularity of definition.

33. There are other issues about number. I have the symbols 1, 2, 3, … and rules for working with them such as $3+1=4$ and that $9+1=10$. I can specify the rules without circularity and so learn to use the numbers. It takes discussion, teaching and learning to do arithmetic. Later, once one has learned to do arithmetic, it is possible to make definitions and theories and worry about circularity. The worry about circularity has led mathematicians to the remarkable notion of *undefined terms*. In every mathematical theory there are terms that are simply not defined. In this way these terms avoid circular definition.

34. Let us go back to the standard for 2. This is built from sets, and sets are, in their own theory, undefined, but have special operations for building them. The simplest set is the empty set, usually written as { }. This set is distinguished by the fact that it has no members, and is represented as an empty container. As a container, it is a distinction with no contents and no markings other than its boundary. I cannot define a distinction, but I use distinctions all the time and mathematics is built from them. When I go into the realm of distinctions I go into the realm of my own creations and into the realm where I can make the imaginary act that distinguishes myself from what I do. I am a reflexive domain.

The definition of cybernetics and the cybernetics of definition

35. I assert that cybernetics *cannot be defined without circularity*. Recall the theorem of the previous section:

Theorem 1. There is no definition of the concept of distinction that is not circular.

36. I shall use this theorem to prove the next theorem. The point is that cybernetics is so deeply involved in the concept of distinction that it can no more be defined than can distinction be defined without circularity.

Theorem 2. There is no definition of cybernetics that is not circular.

Proof. Cybernetics includes within its purvey the concept of distinction since it studies circular processes and all distinctions are circular processes involving the production of an observer in relation to an observing system and the production of an observing system in relation to an observer. Therefore a definition of cybernetics would entail a definition of distinction. But by Theorem 1 (see above) there can be no definition of distinction that is not circular and therefore there can be no definition of cybernetics that is not circular. *QED*

37. In proving this theorem, I have assumed that cybernetics deals with the fundamentals of circular processes. This is in accord with its history and with its use. Contrast this situation with mathematics. Mathematics studies distinctions, but I do not insist that distinctions themselves are the subject matter of mathematics. Nevertheless, if pressed, I would argue that mathematics is also indefinable!

38. I conclude that a definition of cybernetics could be given in the following form:

Cybernetics is the study of systems and processes that interact with themselves and produce themselves from themselves. This includes cybernetics itself as a system or process that interacts with itself and produces itself from itself.

39. While the reader may object to this definition on grounds that it is not broad enough for a definition of cybernetics, at least the definition is circular.

Cybernetics and reflexive domains

40. I should note that once cybernetics is defined in terms of itself, it becomes what is commonly called “second-order cybernetics.” Perhaps there never was a first-order cybernetics but this is a useful distinction. When I distance myself from an object of study,

but nevertheless examine its circularities, it may be natural to call such a study cybernetics of the first-order. Here I am concerned with the second-order, but I will just call that cybernetics.

41. There is a way to do mathematics that is explicitly circular, called lambda calculus (Barendregt 1985; Kauffman & Buliga 2014; Kauffman 2015; Scott 1971). It is called "Laws of Form" (Spencer-Brown 1969) in its full simplicity.

42. First I want to discuss Laws of Form and then pass over to more complex versions.

43. In George Spencer-Brown's book *Laws of Form* (Spencer-Brown 1969), a very simple mathematical system is constructed on the basis of a single sign, the mark, designated by a circle or a box or a right angle bracket. I shall not develop the formalism here. But that sign, in our eyes, makes a distinction in plane in which it is drawn. And the interpretation of Spencer-Brown's calculus has us understand that the sign refers to the distinction that the sign makes (we make that distinction when we are identified with the sign). Thus the sign of distinction in the calculus of Spencer-Brown is self-referential. That is, expressions in the calculus refer to themselves and to other expressions in the calculus.

44. While this calculus is circular and self-referential, it is consistent.

45. By a series of departures and constructions, the calculus of indications of Spencer-Brown can be seen to be a foundation for the construction of all of mathematics, and it gives direct insight into the structure of language and communication. The external observer is intimately involved in this calculus, for it is through that observer that the distinctions take place. Thus the circularity of the calculus engulfs the observer. One is in a cybernetic domain.

46. A more complex form of circularity in mathematics occurs when an expression *explicitly* refers to itself. For example, I could write the self-referring equation $x = 1 + 1/x$ and have an expression x that is defined in terms of itself. More general fixed points indicate other circularities. For example

"This sentence has thirty-three letters."

is a sentence that refers to itself and tells a truth about itself. I can formalize circularity in terms of fixed points. Heinz von Foerster (2003b) was fond of this relationship of circularity and fixed points. He saw that a fixed point, taken in the cognitive world, can connote an object in the perception of an observer. In his terms, the object is a token for an eigenbehavior, the iteration of a transformation whose fixed point is the object. The observer embodies a transformation T that leaves the object fixed in the sense that one has the fixed point equation $T(\text{Object}) = \text{Object}$. This means that no extra distinction is introduced by the application of the transformation T to the Object. Von Foerster called the object so fixed an *eigenform,* and the transformation an *eigenbehavior*.

47. It should be understood that the object, the eigenform, may come into existence through the action of the transformation. *This can happen by an interlock of meanings without any iteration.* By an interlock I mean a direct reference of the sentence to itself, or a mutual reference of two structures. For example, the US dollar bill has a statement written upon it that says "This bill is legal tender." That statement interlocks with the dollar bill upon which it is printed. When we shake hands, each hand grasps the other hand. The hands interlock. In the next example, we have the sentence "I am the one who says I." The sentence prior to the one you are now reading shakes hands with its reader and makes an opportunity for the reader to identify himself or herself.

48. I have the following operator:

$$T(x) = \text{the one who says } x$$

The eigenform for this operator is "I".

"I am the one who says I"
"I = the one who says I"
$$I = T(I)$$

49. This explication of "I" as an eigenform places its fundamental circularity starkly before us. I is an eigenform of the transformation, but the identity of the I can vary. I can be the one who says I and so can you. The eigenform I is fixed by a linguistic transformation but is handed back and forth in the conversation among a community of I's. This is exactly the condition that Spencer-Brown (1969: 76) describes for his mark when he says "We see now that the first distinction, the mark and the observer are not only interchangeable, but in the form, identical."

50. A community of observer/participators forms a *reflexive domain D*. By this term I mean that each person in the domain is also an actor in that domain. Each one acts upon the others and each can be acted upon by the others and by himself. This is the fundamental condition for the cybernetics of social and economic behavior, and it is also the condition for all scientific behavior.

51. There is a key relationship between reflexive domains and eigenforms. I state this relationship as a theorem.

Church-Curry fixed point theorem. Let D be a reflexive domain and let T be any element of the domain D. Then there is an element X in D such that X is a fixed point for T. That is, $TX = X$.

Proof. Define the following operation on elements of D:

$$Gx = T(xx).$$

That is, G operates on x by letting T operate on the result of x operating on itself.
Since D is a reflexive domain, I can assume that the operation G is itself an element of the domain. Now examine the result of G operating on itself. G operating on itself is the result of T operating on G operating on itself. That is, $GG = T(GG)$. Thus GG is a fixed point for T. *QED*

52. It is quite possible that the reader will find this construction, simple as it is, quite baffling! It is best to discuss it from a number of angles. The key point for a reflexive domain is that participants in the domain can act on themselves. One must take this property to heart. Regard yourself as such a participant and understand that you interact not only with others (for example in a market) but also with yourself. There is a singularity about interacting with oneself that is different from interacting with others. You have to separate yourself from yourself to interact with yourself and this separation is at a conceptual level or an imaginary level, since you are not actually separate from yourself. Think of the G in the theorem as a person. This person acts on another person P by letting the transformation T act on the self-interaction of P. What does it mean to act on the self-interaction of another? It means to go into conversation with that other. Thus G induces T to go into

conversation with P. One can think of G as a coach who gets people to interact with one another. But how does G act on himself?

53. Well G acting on G induces T to be in conversation with G. This means that G's self-interaction is acted on by T. Thus G's self-interaction is indeed the action of T on G's self-interaction. The singularity of self-interaction leads to the fixed point.

54. This singularity of self-interaction is well-known in many puzzles and paradoxes. For example, there is the tale of the village with a barber B who shaves only those who do not shave themselves. This is straightforward just so long as I do not ask "Who shaves the barber?" For if the barber shaves himself then by definition, he cannot shave himself.

55. A deeper relationship with the fixed point theorem is Russell's paradox. For this purpose, let AB denote the relationship "B is a member of A." Thus A acting on B is the act of B becoming a member of A. Then I can define $Rx = \sim(xx)$ where $\sim$ denotes "not." R is a set with the property that *x is a member of R exactly when x is not a member of x.* R is the set of all sets that are not members of themselves. This set R is the famous Russell set. The paradox of the Russell set is that

56. RR is an eigenform for negation. For I have

$$Rx = \sim(xx) \text{ and so } RR = \sim(RR).$$

57. This Russellian eigenform is a problem for those who insist that negation must not have fixed points. Certainly RR is a rogue set from the usual point of view since it apparently can both belong to itself and not belong to itself.

58. This depends upon what is the "usual point of view." Looking from the reflexive domain of persons and selves, I see that RR is very like a person. RR is part of itself, but RR is also the whole of herself and so not a part at all. RR is a member of RR and RR is not a member of RR. In the reflexive domain of persons and selves it is natural enough for RR to be a eigenform of negation.

59. It is an axiom of physical science that one requires results and procedures that can be repeated by others to a sufficient degree of accuracy so that the community of physicists agrees that the results and procedures do not depend on the specific observers (other than that they carry out the actions of the experiment according to the specifications). It should be clear from this well-known description of experimental physical science that an experiment is a transformation E that involves persons P and a result X and I desire an eigenform X so that $E(P,X) = X$ independent of the choice of P (within a community of "competent scientists" P). Why does X appear on the left-hand side of this equation? By this I mean that the experiment is repeatable. You can record X as a result of the experiment and then find that when you perform it again, you get the same result. It is the repeatability that makes a successful experiment into an eigenform.

60. That knowledge should be independent of the observers is related to the repeatability and I will discuss it further below.

61. Physics is defined circularly and it demands certain eigenforms for its successful experiments. These eigenforms are in accord with the description of a world that is objective, a world that is independent of the particular choice of observers. The objective world of physics is not given a priori. The objective world of physics is a construction of the physicists based on the type of results that they deem acceptable as valid physics. Of course it could have happened that physics would fail and there would never be any experiments E of this kind or any results X of this kind. Fortunately for physics, there is a whole world of physical results, and it would appear that there is no end in sight to the progress of physics.

62. I see that one could look at a science that studies eigenforms and reflexive domains and that the participants in such cybernetics are themselves elements/actors in the reflexive domain. One is then led to the question: Which is more fundamental, the circularity of eigenform or the reflexivity of the domains in which these eigenforms occur?

63. It is easier to start with eigenform since I have a sense about how to make an eigenform by iterating a transformation. When one does this iteration one learns curious lessons.

64. One lesson is the astonishing fact that one can make an eigenform just from the form of a transformation as in $X=TTTT\ldots$. Here, X is an eigenform for T since formally $TX=X$. But is this not some sort of magic trick, a sleight of hand? Indeed it is. For I could define

$$T^1(\mathrm{g})=T(\mathrm{g})$$
$$T^2(g)=TT(g)$$
$$T^3(g)=TTT(g)$$
$$\ldots$$
$$T^N(g)=TTTT\ldots T(g)$$

where there are N T's in this composition. Then $T^N(g)$ depends upon g as well as N, for each N. I can imagine $TT\ldots TTg$ with an infinite number of T's in-between the first and the last T, and this would be an eigenform that depended on the choice of g. I would have to imagine an infinity where I can "count down" infinitely as well as counting up.

65. The simplest way to think of this is to imagine that N is a generic large integer so that it is possible to confuse N and $N+1$. Then N acts as though it is infinite, but is actually finite but large.

66. Our usual method produces an infinite limit

$$X=TTTT\ldots \text{ with } TX=X.$$

67. The fixed point does not depend upon the initial conditions. If I take an example such as

$$Tx=1+1/x,$$

then $X=TTT\ldots=1+1/(1+1/(1+\ldots))$ and $X=1+1/X$. I can ask for numbers that satisfy this equation, and there are numerical solutions (the golden ratio, phi$=(1+\mathrm{sqrt}(5))/2$ is a solution to this equation.). This independence of the eigenform on the initial conditions is analogous to an experiment whose results are independent of the particularities of the experimenter.

68. I would like to make a more precise version of this analogy with the possible independence of observation so that one could consider experiments that are both sensitive and insensitive to the conditions of the experimenter. I describe a scientist to be in search of eigenforms that are independent of the action of the observer. Iteration of a transformation can, as explained in the previous paragraph, erase some of the initial conditions that are characteristic of a particular observer. This is only part of the story. To make a complete account of how a given science strives to make its results observer-independent is a project for future workers in this field of second-order science.

69. Eigenforms occur within a reflexive domain, giving rise to recursion, and recursion itself brings forth that reflexive domain. The entire reflexive domain is an eigen-

form. This shows how eigenforms can be dynamic. The entire world of science is itself an eigenform. The world of the stock market is an eigenform. The cybernetics world is an eigenform. These eigenforms are the forms of closure that occur in large interacting systems where the dynamics of individuals leads their identities outward and the individuals become identified with their actions. Just so do words become identified with their actions in Ludwig Wittgenstein (1958: §43): "The meaning of a word is its use […]." In the same way, Charles Sanders Pierce (Kauffman 2001) had the concept of the individual as a sign for himself. This means that in the dynamics of sign-behavior there arises reference and eventually self-reference and through this self-reference the individual becomes a sign for himself. Word and individuals are both seen as elements of a reflexive domain in the work of Wittgenstein and Pierce. Pierce's concept is directly related to the reflexive domain in which the individual comes to have an identity through the eigenform "I".

70. With this I return to the identification of the mark and the observer in Spencer-Brown (1969), now at the scale of the detailed interactions of individuals with many others.

71. I have the fundamental theorem that every element T of a reflexive domain D has a fixed point X with $T(X)=X$. Eigenforms prevail at the individual as well as the global levels in reflexive domains. The world of reflexive domains and their eigenforms is the world in which cybernetics occurs.

72. How does a reflexive domain come about? In some cases, such as our personhood, one simply lives there and cannot easily give an account of how it happened. I live in the eigenform of my reflexive domain of self-other interaction. It is a mystery how it happens. I do not have a sequential construction for it. On the other hand, consider some given domain D that is not reflexive. By not reflexive I mean that the transformations of D to itself are not in perfect correspondence to elements of D. There are transformations A taking D to D that are not members of D. What should I do? I may include them and form a new domain D^1 But D^1, while larger than D, may still not be reflexive, so I may need to adjoin more elements. The general pattern is $T(D^n)=[D^n, D^n]=D^{(n+1)}$. Then I iterate the construction of enlarging the domain to infinity or to a very large N where D^N and $D^{(N+1)}$ are indistinguishable. At this point I have achieved D^N as an eigenform for T. I have created a reflexive domain. Once I have a reflexive domain, I can avail myself of the Church-Curry fixed point theorem within that domain and all the other amenities of closure that such domains afford. The insight that reflexive domains are themselves eigenforms is directly related to how they can come into existence. The technical details of this limiting process can be seen in the work of Dana Scott (1971).

73. With this point of view, I can think of a reflexive domain as arising in the freedom to assign names to processes, and to allow those processes to become new elements of the original domain of objects (under scrutiny). This sort of freedom is well-known to computer programmers of languages, where an algorithm or procedure is given a name and that name is then at the same level as other original terms in the language. The language expands in time as the programmer makes new "macros" of this sort. In this practical situation one does not have to go to the limit of an infinite tower construction (as described in the previous paragraph) to obtain reflexivity. One only has to recognize that defined processes are subject to being elements of an always-extending language.

74. The construction I made in the Church-Curry fixed point theorem can be accomplished under relatively ordinary circumstances. For example, suppose that I define $Gx=<xx>$ as a *symbolic process*. It is a process that a computer can perform on symbol strings such as x. Then applying G to x means that I duplicate x and place brackets around

the two copies of x. This is an operation that can be performed by a computer. I regard G itself as a string of symbols (with one character). The strange and curious thing is what happens when I apply G to G.

75. With $Gx=<xx>$, I obtain $GG=<GG>$. I obtain a statement of self-reference and, if I take this equality seriously as an action, I obtain the unending recursion (GG is replaced by $<GG>$)

$$GG=<GG>=<<GG>>=<<<GG>>>$$
$$=<<<<GG>>>>=\ldots.$$

76. All this is compatible with the previous notion that

$$E=<<<<<<<\ldots>>>>>>>$$

is invariant under bracketing: $E=<E>$.

77. In one case, I obtain the eigenform by singular substitution. In the other case, I obtain the eigenform by a limiting process. It should be noted that these examples of strict substitution and transition to infinity are idealized and do not begin the wonderful partial aspects of self-reference that occur in our worlds of language. For example, I may refer to myself without yet knowing what boundaries of the self are intended and indeed with the hope of extending them. I may say "I am a mathematician" or "I am a clarinetist," in each case using the declaration as a performative act that may promote the one who says it into that category. The act of saying can be an essential ingredient in the movement to a state of being. Thus does one say "With this ring I thee wed."

78. It is at this point that the reader will discern that I am in favor of both nouns and verbs. In fact, from our point of view it is indispensible to have both nouns and verbs, and the possibility of changing nouns to verbs and verbs to nouns as the need arises. Due to the long discussions in systems theory about the nature of nouns and verbs, this aspect of our epistemology will be continued in other papers.

79. I shall now see how self-reference can arise in systems that are powerful enough to have reference, and powerful enough to take the names of processes and the names of things and use them in the language on an equal basis (just as one does in human natural languages). Nouns and verbs become interchangeable. Suppose that I have a system S that can make distinctions in its world and give names to the sides of these distinctions.

80. S has a language, and in that language S will write $A\rightarrow B$ to mean "A refers to B" or that "A is the name of B." S is endowed with a special process denoted by "#" so that when $A\rightarrow B$, S creates a new reference $\#A\rightarrow BA$ called the *indicative shift* of the original reference $A\rightarrow B$. Thus $\#A$ is the name of "B with a name-tag A attached."

81. I say that $\#A$ is the meta-name of BA. This shifting of names to meta-names is fundamental to the observing system S. The shift enables an object and its name to be superimposed just as I find that persons I know have their name as one of their direct properties for me. It is the shift of naming that allows the system S to achieve self-reference. The system S will create a name for the operation of the shift itself.

82. Let $M\rightarrow\#$ denote the name of the shift operation. Note that this is an instance of a process (#) acquiring a name (M) and thereby being associated with a noun. But then this naming of the shift process can itself be shifted, and I find that $\#M\rightarrow\#M$ is the result of that shift. In this way, the system S acquires self-reference. If I let $I=\#M$ then I find that I refers to I. If the system could talk, she would say "I am the meta-name of my own meta-naming process!" The structure of this acquisition of self-reference has the same

pattern as the Church-Curry fixed point theorem. Our point in describing it is to show that the self-reference of "I am the one who says I." occurs naturally in a model of a system *S* that is capable of reference, naming and distinction. Such systems deserve the name of "observing systems."

Conclusion

83. I am now in a position to discuss second-order science in the light of an understanding of reflexive domains. I have discussed how a society of persons can be schematically modeled by a reflexive domain *D*. The domain contains more than just the human persons. It contains processes that are initiated by them, with each process regarded both as an object in the domain and as a transformation of the domain. The transformative properties of an element can be referred to any other element or person in the domain. I can now look at the practice of science in terms of the landscape of the domain *D*. I emphasize that the domain *D* should be seen as "the world" in the sense of Wittgenstein. Here I refer to a world that begins in the context of the early Wittgenstein of the *Tractatus* (Wittgenstein 1922) but continues into the world of the later Wittgenstein of the *Philosophical Investigations* (Wittgenstein 1958). From the point of view of reflexive domains these worlds are not incompatible. What is the case is the present state of action and eigenform. But there is no longer an impenetrable barrier between the descriptions (pictures) of the world and the world itself. Each acts upon the other. Each creates the other. Words acquire their meanings via actions, and distinctions occur in the sharp invariances of certain eigenforms. The world is everything that is the case, and the world evolves according to the theories and actions of the participants in that world. Thus when a scientist in the world proposes a theory, this theory becomes a new element of that world. The theory acts and is acted upon by the world of persons and elements of the world. There is no protection for the world from the effects of such a theory. It is remarkable that physicists have been able to isolate phenomena that do not appear to be affected by the theories of those phenomena.

84. For example, the Maxwell theory of electromagnetism does not appear to affect the behavior of electromagnetic fields. What has been discovered is that the electromagnetic field (as measured by the procedures associated with this physics) is an eigenform of the theory. The physics is defined to be physics when the object of the theory is found to be an eigenform of the theory. But I can take a wider view and examine the effect of the electromagnetic theory in the world at large, not just its effect on the "fields themselves."

85. Since James Clerk Maxwell and the allied discoveries of electrical generators and electromagnetic wave transmission, the entire face of our world has been transformed by the entry of the theory of electromagnetism into that world. This is so striking that it seems hard to believe that it is a common belief that scientific theories are just objective descriptions of the way the world "is." The way the world "is" is an evolving context for interaction and exploration.

86. The key to the apparent objectivity of a science is in its carefully crafted eigenforms that are checked to remain fixed throughout significant periods of time. If one attempts to maintain this sort of objectivity in other forms of social science, the processes can take a much different shape. For example, one can design trading algorithms for the stock market on the basis of seemingly reliable information about the behavior of the market. But these very algorithms, when entered into daily practice in the market itself affect the action

of the market and can even lead to instabilities and behaviors that were nowhere in the original theories. Certainly the same remarks apply to the theories and even the opinions of investors, which when made public can affect the action of the market itself. One may say that this indicates that the market is not a subject for objective science, but this is not necessarily the case. Our definition of objective science is that the actions in the reflexive domain produce relatively stable eigenforms. Thus a new version of stock market economics could arise that searches for such regularities even in the face of the publication of the theories themselves.

87. Other aspects of knowledge certainly have this reflexive pattern. There is a science to learning to play a musical instrument, and it involves principles that do not appear to be changed by the acts of practice. But other aspects such as learning to improvise are clearly a matter of finding a moving eigenform in the course of the action of the play. A simpler example is learning to ride a bicycle. The riding of a bicycle is an eigenform in action and there is no way to find this eigenform except to perform the action. Theories of games that do not submit to exact analysis such as chess or Go evolve in relation to the play of the game. The known theories of chess have seriously affected tournament play, and in turn this tournament play has changed the evolution of the theories.

88. Here is a remarkable story told by the physicist John Archibald Wheeler about a game of twenty questions:

> Then my turn came […] I was locked out an unbelievably long time. On finally being readmitted, I found a smile on everyone's face, a sign of a joke or a plot. I nevertheless started my attempt to find the word. 'Is it an animal?' 'No.' Is it a mineral?' 'Yes.' 'Is it green?' 'No.' 'Is it white?' 'Yes.' These answers came quickly. Then the questions took longer in the answering. All I wanted from my friends was a simple 'yes' or 'no.' Yet the one queried would think and think before responding. Finally I felt I was getting hot on the trail, that the word might be *cloud*. I knew I was allowed only one chance at the final word. I ventured it: 'Is it *cloud*?' 'Yes,' came the reply, and everyone in the room burst out laughing. They explained to me that there had been no word in the room. They had agreed not to agree on a word. Each one questioned could answer as he pleased – with one requirement that he should have a word in mind compatible with his own response and all that had gone before. Otherwise, if I challenged, he lost. This surprise version of Twenty Questions was therefore as difficult for my colleagues as it was for me […] What is the symbolism of the story? The world, we once believed, exists *out there* independent of any act of observation. […] I, entering the room, thought the room contained a definite word. In actuality, the word was developed step by step through the questions I raised […] Had I asked different questions or the same questions in a different order I would have ended up with a different word […] However, the power I had in bringing the particular word *cloud* into being was partial only. A major part of the selection lay in the 'yes' or 'no' replies of the colleagues around the room […] In the game, no word is a word until that word is promoted to reality by the choice of questions asked and answers given. (Davies & Brown 1986: 23f)

Wheeler's allegorical fable was intended to illuminate the conditions of the quantum physicist. In quantum physics, no phenomenon is an actual phenomenon until it is observed and agreed upon by all the physics colleagues. The story just as well illustrates the world of social interaction.

89. My thesis is that all attempts to find stable knowledge of the world are attempts to find theories accompanied by eigenforms in the actual reflexivity of the world into which one is thrown. The world itself is affected by the actions of its participants at all levels. One finds out about the nature of the world by acting upon it. The distinctions one makes change

and create the world. The world makes those possibilities for distinctions available in terms of our actions. Given this point of view, one can ask, as one should of a theory, whether there is empirical evidence for this idea that stable knowledge is equivalent to the production of eigenforms. In this case we have only to look at what we do and see that whenever "something is the case" then there is an orchestration of actions that leaves the something invariant, making that something into an eigenform for those actions. The eigenform thesis is not itself a matter of empirical science. It is a matter of definition, albeit circular definition. Another point of view is that the empirical evidence is all around you. Examine any thing. How does it come to be for you? Investigate the question and you will find that thing is maintained by actions. The action could be as simple as opening your eyes and looking at the cloudy sky. With that action, the cloudy sky comes to be for you. I do not assert that this is the usual scientific explanation of cloudy sky. But if you want to work with such things then it is usually even more transparent. The sharp spectral lines of helium are the result of setting up a very particular experiment that produces them. The experiment, its equipment, the scientists and all that is needed to perform it is the transformation whose eigenform is the spectrum of helium.

90. It is a fruitful beginning to look at present scientific endeavors and to see how they are interrelated and find connections among them, to engage in meta-scientific activity. This can reveal how theories, seemingly objective, actually affect the world through their very being, and how these actions on the world come to affect the theories themselves. In exploring the world, we find regularities. It is possible that these regularities are our own footprint. In the end we shall begin to understand the mystery of the eigenforms that we have created, constructed and found.

Open Peer Commentary:
Remarks from a Continental Philosophy Point of View

Tatjana Schönwälder-Kuntze

1. For me, the great merit of Louis Kauffman's target article is that it brings together – intentionally or not – abstract mathematical and logical formalism with philosophical reflections ranging from everyday-life experience to scientific experience through second-order cybernetics. The article tries to support two interlocked theses within a mathematical framework. The first states that scientific observers always observe themselves, at least in some way. The second says that the observed objects depend on their observers, insofar as they are shaped by the observation (§4). From my point of view, there is nothing to reject. Maybe one could add that the observers are also shaped by the forms and categories with which they observe their objects and subsequently by the knowledge about them – but this is another discussion I do not want to enter into here. Instead, I want to draw attention to some striking similarities between "classical" philosophical approaches, Wittgenstein aside, and the statements in Kauffman's article.

2. Being trained in philosophy of science and formal logic, and being especially familiar with George Spencer Brown's *Laws of Form* (1969), but thinking and doing research in the field of continental philosophy, I want to make two remarks: